MIL-P-50308B(MU)
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SUPERSEDING
MIL-P-50308A(MU)
8 SEPTEMBER 1976

MILITARY SPECIFICATION

PROJECTILE, HEAT, 106MM, M344A1 WITH FIN M8
METAL PARTS ASSEMBLY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification is the applicable detail specification for Projectile HEAT, 106MM, M344A1 with Fin M8, Metal Parts Assembly.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue on effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military
MIL-A-2550 - Ammunition, General Specification For

STANDARDS

Military
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-1167 - Ammunition Data Card
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1169 - Packaging, Packing and Marking of Inert Ammunition

FSC 1315

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Frankford Arsenal, ATTN: SARFA-MDM, Philadelphia, PA 19137 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2.1 (Cont'd)

MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

Armament Command

F7549103	- Projectile, HEAT, 106MM, M344A1, Metal Parts Assembly
D7549111	- Fin Assembly, M8
F8837835	- Pallet for Fixed and Semi Fixed Shell
F7548319	- Cartons, Ammunition, for Interplant Shipment of Projectiles Empty
LI-7549103	- Inspection Equipment List for Projectile, HEAT M344A1 Metal Parts Assembly
LI-7549111	- Inspection Equipment List for Fin Assembly, M8

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity of as directed by the contracting officer.)

2.2 Other publications.-The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ANSI-B46.1 - Surface Texture

(Application for copies should be addressed to the American National Standards Institute (ANSI), 1430 Broadway, New York, NY 10018.)

ASTM E-8	- Tension Testing of Metallic Materials
ASTM B-117	- Salt Spray (Fog) Testing

(Application for copies should be addressed to the American Society for Testing and Materials, 916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 General.-Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings, referenced specifications and the following:

3.2 Heat treatment of stress relief.-The heat treatment or stress relief applied to any component shall be uniform throughout and shall be so conducted as to produce in each component the mechanical properties specified on the applicable drawing.

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3.3 Wire assembly.

3.3.1 Pull test.-The wire eyelet assembly shall withstand a pull test of 90 pounds, min.

3.3.2 Electrical continuity.-The wire assembly when assembled to the metal parts assembly shall exhibit no evidence of electrical discontinuity between the terminal eyelet and the fuze wire terminal.

3.3.3 Fuze wire terminal.-The fuze wire terminal when crimped to the fuze shall withstand a pull test of 20 pounds, min.

3.3.4 Insulation breakdown.-The wire assembly when assembled to the metal parts assembly, shall exhibit no evidence of electrical breakdown when subjected to a direct current potential differential of 1500 volts.

3.4 Electical resistance.-After assembly, the resistance between the forward end of the ogive and the rear of the body shall not exceed 0.2 ohms, max.

3.5 Protective coating.-Protective coating shall be in accordance with the applicable drawings and specifications. Where protective coating is required by component details, finish requirements will apply to all indicated surfaces of the component regardless of contour configuration.

3.6 Painting.-Painting shall comply with the requirements of the applicable drawing. All paint and marking shall be dry to the touch before packing for shipment. Drying time prior to testing shall be in accordance with the applicable specification.

3.7 Metal defects.

3.7.1 Components.-The components including the cavity shall be free from cracks, splits, burst, cold shuts, pipe, porosity, inclusions, folds, seams, and other metal defects.

3.7.2 Components.-The cavity of the components shall be free of scale, fins, burrs, draw marks, laminations, imbedded foreign matter, pits, and sharp edges.

3.7.3 Visual standards.-Visual standards for metal defects of paragraph 3.7.2 will be established for each contract after the first months production or whenever requested by the procuring contracting agency. (See 6.2)

3.8 Proving ground functional test.-Projectile assemblies, when tested at excess and service pressures, shall show no evidence of metal parts failure. Metal parts failure shall be:

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3.8 (Cont'd)

a. Loss or break-up of metal parts components in flight.

b. Fin blades not opening when fired.

c. Not capable of defeating 6 inches of homogeneous armor plate at 60 degrees obliquity at all useable ranges.

3.9 Workmanship.-The requirements for workmanship are as specified on the applicable drawings, specifications and the following:

3.9.1 Material defects.-Molded components shall be free of porosity and excessive flash.

3.9.2 Hydrogen embrittlement.-Copper cones which are annealed during manufacture shall not show gassing or open grain structure characteristics of hydrogen embrittlement.

3.9.3 Burr.-No component shall have a burr which might interfere with the assembly of the projectile or which might be injurous to personnel handling the item.

3.9.4 Foreign matter.-All parts and assemblies shall be free of chips, dirt, oil, rust, grease, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspection set forth in the specification where such inspections are deemed necessary to insure supplies and services conform to the prescribed requirements.

4.1.1 Quality assurance terms and definitions.-Reference shall be made to Standard MIL-STD-109 to define Quality Assurance terms used.

4.1.2 Inspection.-Inspection shall be in accordance with Specification MIL-A-2550 except as specified herein.

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4.1.3 Inspection criteria.-

a. Successful passage of all inspection required by the drawings and specifications shall be considered as a condition precedent to performing the proving ground functional tests. (See 3.8)

b. Acceptance or rejection of a lot by the Government shall be contingent on successful passage of the proving ground functional tests.

4.2 First article sample.

4.2.1 Initial production sample.-At the beginning of regular production a sample shall be submitted in accordance with contract requirements and shall consist of 50 metal parts assemblies with Fin M8 as follows:

35 metal parts assemblies complete, painted
5 metal parts assemblies complete, unpainted
5 sets of subassemblies
5 sets of component parts, unassembled

Thrity of the above painted metal parts assemblies shall be forwarded to a designated proving ground for ballistic testing. The sample shall be manufactured using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging and packing shall be the same as used in regular production and shall be obtained from the same source of supply.

4.3 Inspection.

4.3.1 Lot.

4.3.1.1 Submission of product.-The product shall be submitted in accordance with MIL-STD-105.

4.3.1.2 Cone lot.-A lot of cones shall contain cones from not more than one lot of copper discs of uniform size and temper.

4.3.1.3 Heat treatment lots.-Heat treatment lots shall contain components from not more than one heat of steel that have been heat treated in the same heat treatment equipment, at the same temperature, in one unchanged process. The heat treatment lot thus constituted shall be the basis for the selecting the samples for the mechanical properties test hereinafter prescribed. (See 6.7)

4.3.1.4 Stress relieved lots.-Stress relieved lots shall contain components from not more than one heat of steel that have been cold shaped by the same process and stress relieved in the same furnace at the same temperature for the same length of time. The stress relieved lot thus constituted shall be the basis for selecting the samples for the mechanical properties test hereinafter prescribed.

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4.3.1.5 Projectile lots.-When it has been determined that bodies of a heat treated or stress relieved lot have complied with the mechanical properties requirement of the applicable drawing, they may be grouped for further processing.

4.3.1.6 Shipping lots.-Shipping lots shall consist of projectile assemblies produced by one supplier, under one contract, in one unchanged process, in accordance with the same drawings, revisions, under conditions of a continuous production, free of interruptions other than those due to the end of shift day, or work week. Shipping lots shall be assigned a lot number in accordance with MIL-STD-1168. The shipping lot thus constituted shall be the basis for selection of samples for the proving ground functional test. Lot size shall be determined by the procuring contracting officer.

4.3.2. Examination.-One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on an individual defect basis in accordance with the classification of defects contained herein, using sampling plans contained in MIL-STD-105. Continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the procuring activity. An AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect shall be used. All nonconforming material shall be rejected. The roughness comparison specimens prescribed by ANSI-B46.1 shall be used as a basis for surface roughness determination. Four power (4X) magnification may be used in performing visual examination.

4.3.2.1 Tool control dimensions.-

a. Dimensions marked tool control shall be gaged at the beginning or end of each working day (24 hrs), during First Article Inspection and for the first component after each tooling change, adjustment or repair. Where destruction of the component is necessary to gage such dimensions, measurements of the tool may be substituted provided the contractor has established correlation between the tool dimensions and the component dimensions prior to the start of production.

b. When the contractor has satisfied (a) above, visual inspection of tool control dimensions will be permitted. Visual inspection shall be performed every two (2) hours for each machine in operation. If any tool control dimension(s) is in doubt during visual inspection, the dimension(s) shall be immediately gaged (see "a" above) to determine acceptablility. Corrective action will immediately be instituted by the contractor whenever a defective tool control dimension is found.

4.3.2.2 Cap, ogive, Drawing C7549082, covering a detail of Drawing F7549103.

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4.3.2.2 (Cont'd)

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. Runout at front with pitch diameter of thread and rear face	Gage
102. Pitch diameter of thread, max.	Gage
103. Minor diameter of thread, max.	Gage
104. Diameter at front	Gage
105. Diameter at rear, min.	Gage
106. Distance between rear of external groove and front of lucky assembly cavity	Gage
107. Depth from rear face to front of lucky assembly cavity	Gage
108. Thickness through drill point location, min.	Gage
109. Thickness of wall in lucky assembly cavity	Gage
110. Metal defective	Visual
MINOR	
201. Diameter of external groove	Gage
202. Diameter of small cavity (to drill point)	Gage
203. Depth from rear face to start of thread undercut	Gage
204. Depth from rear face to start of drill point in small cavity	Gage
205. Length, total	Gage
206. Width of external groove	Visual (See 6.1)
207. Radii or chamfers missing or incorrect	Visual (See 6.1)
208. Surface finish, improper	Visual (See 6.1)
209. Protective coating inadequate ot damaged	Gage
210. Burr	Gage
211. Foreign matter	Visual

4.3.2.3 Wire assembly, Drawing C7549090, covering a detail of Drawing F7549103

CRITICAL

None defined

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4.3.2.3 (Cont'd)

	Defect	Method of Inspection
MAJOR		
101.	Diameter of eyelet, inside, min.	Gage
102.	Insecure assembly	Visual/Manual
103.	Terminal cracked or bent	Visual
104.	Component damaged	Visual
MINOR		
201.	Foreign matter	Visual

4.3.2.4 Ogive, Drawing D7549078, covering a detail of Drawing F7549103.

CRITICAL		
1.	Metal defective	Visual (Probe Light)
MAJOR		
101.	Runout of pitch diameter of small thread with pitch diameter of large thread , cap seat and rear face	Gage
102.	Pitch diameter of small thread, min.	Gage
103.	Major diameter of small thread, min.	Gage
104.	Pitch diameter of large thread, min.	Gage
105.	Major diameter of large thread, min.	Gage
106.	Diameter wire assembly cavity	Gage
107.	Diameter, large, outside	Gage
108.	Diameter of cone seat, max.	Gage
109.	Diameter of cone seat groove	Gage
110.	Diameter at rear of large cavity	Gage
111.	Depth of wire assembly cavity	Gage
112.	Depth from rear face to front of cone seat groove	Gage
113.	Length from front face to cap seat	Gage
114.	Length from rear face to forward basic diameter	Gage
115.	Length from rear face to read basic diameter	Gage
116.	Length from rear face to large outside diameter	Gage
117.	Length, total	Gage
118.	Thickness of wall	Gage
119.	Width of cone seat groove	Gage
MINOR		
201.	Diameter, small thread undercut	Gage
202.	Diameter, large thread undercut	Gage

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4.3.2.4 (Cont'd)

	Defect	Method of Inspection
203.	Diameter of chamfer at rear of cone seat	Gage
204.	Depth from rear face to base of ogive cavity	Gage
205.	Width, small thread undercut	Gage
206.	Width, large thread undercut	Gage
207.	Presence and location of wrench holes	Visual (See 6.1)
208.	Radii or chamfer, missing or incorrect	Visual (See 6.1)
209.	Surface finish, improper	Visual (See 6.1)
210.	Paint on other than indicated interior surfaces	Visual
211.	Interior protective coating, inadequate	Visual
212.	Burr	Visual
213.	Foreign matter	Visual

4.3.2.5 Body, Drawing D7549077, covering a detail of Drawing F7549103.

CRITICAL

1.	Metal defective	Visual (Probe Light)

MAJOR

101.	Runout of pitch diameter of rear thread with bourrelets and chamber seat	Gage
102.	Runout of charge cavity with bourrelets	Gage
103.	Runout of rear of body with bourrelets	Gage
104.	True position of rear thread vounterbore with pitch diameter of thread	Gage
105.	Pitch diamter of front thread, max.	Gage
106.	Minor diameter of front thread, max.	Gage
107.	Diameter of bourrelets, min.	Gage
108.	Diameter of front thread undercut	Gage
109.	Diameter of rear thread undercut.	Gage
110.	Diameter counterbore at rear	Gage
111.	Diameter outside at rear of crimp groove	Gage
112.	Depth of crimp groove	Gage
113.	Surface finish of bourrelets, diameter, improper	Visual/Tactile

MINOR

201.	Diameter of charge cavity	Gage
202.	Diameter of body, outside	Gage
203.	Depth from front face to bottom of chamber seat	Gage
204.	Depth from rear face to chamber seat	Gage

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4.3.2.5 (Cont'd)

	Defect	Method of Inspection
205.	Depth of rear thread counterbore	Gage
206.	Length from front face to rear of forward bourrelet	Gage
207.	Length of rear bourrelet	Gage
208.	Length, total	Gage
209.	Location of crimp groove	Gage
210.	Location of rear bourrelet	Gage
211.	Radius at rear of crimp groove	Gage
212.	Width of forward thread undercut, max.	Gage
213.	Width of rear thread undercut, max.	Gage
214.	Presence of slot	Visual (See 6.1)
215.	Radii or chamfers, missing or incorrect	Visual (See 6.1)
216.	Surface finish improper (except as otherwise classified)	Visual (See 6.1)
217.	Paint in cavity inadequate or excessive	Visual
218.	Interior paint on other than indicated surfaces	Visual
219.	Burr	Visual
220.	Foreign matter	Visual

4.3.2.6 Cone, Drawing C7549079, covering a detail of Drawing F7549103.

CRITICAL

None defined

MAJOR

101.	Runout at rear with forward outside diameter and front face	Gage
102.	Rear face not parallel with front face	Gage
103.	Diameter, outside, at front, min. (including ovality)	Gage
104.	Length to forward inside basic diameter	Gage
105.	Length to rear inside basic diameter	Gage
106.	Length to outside basic diameter	Gage
107.	Thickness of wall	Gage
108.	Variation of wall thickness axially	Gage
109.	Variation of wall thickness circumferentially	Gage
110.	Surface finish, improper	Visual (See 6.1)

MINOR

201.	Diameter of conduit hole	Gage
202.	Length, total	Gage
203.	Location of conduit hole	Gage
204.	Thickness through base.	Gage
205.	Radii or chamfers missing or incorrect	Visual (See 6.1)
206.	Burr	Visual
207.	Foreign matter	Visual

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4.3.2.7 Chamber, Drawing F7549080, covering a detail of Drawing F7549103.

Defect	Method of Inspection
CRITICAL	
1. Metal defective	Visual
MAJOR	
101. Runout of external thread undercut with pitch diameter of external thread, and front face	Gage
102. Runout of pitch diameter of large internal thread with pitch diameter of external thread and front and rear faces	Gage
103. Runout of pilot diameter with pitch diamter of large internal thread and rear face	Gage
104. Runout of outside body diameter with pitch diameter of external thread and front face	Gage
105. True position of fuze cavity with pitch diameter of external thread	Gage
106. True position of fuze slots with fuze cavity	Gage
107. Pitch diameter of external thread, min.	Gage
108. Pitch diameter of external thread, max.	Gage
109. Major diameter of external thread, min.	Gage
110. Pitch diameter of small internal thread, max.	Gage
111. Pitch diameter of small internal thread, min.	Gage
112. Minor diameter of small internal thread, max.	Gage
113. Pitch diameter of large internal thread, max.	Gage
114. Pitch diameter of large internal thread, min.	Gage
115. Minor diameter of large internal thread, max.	Gage
116. Diameter of fuze cavity	Gage
117. Diameter of external thread undercut	Gage
118. Diameter of body	Gage
119. Diameter of pilot boss (outside)	Gage
120. Diameter of pilot boss (inside)	Gage
121. Depth of large internal thread counterbore	Gage
122. Length of pilot boss	Gage
123. Angle of dovetail at rear of small internal thread	Visual (see 6.1)

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4.3.2.7 (Cont'd)

Defect		Method of Inspection
MINOR		
201.	Diameter of counterbore in front face	Gage
202.	Diameter of fuze cavity counterbore	Gage
203.	Diameter across fuze slots	Gage
204.	Diameter of small internal thread undercut	Gage
205.	Diameter at rear of dovetail	Gage
206.	Depth of counterbore on front face	Gage
207.	Depth of angle forward of fuze cavity	Gage
208.	Depth from front face to rear of dovetail	Gage
209.	Depth from rear of dovetail to bottom of fuze cavity counterbore	Gage
210.	Depth from front of dovetail to bottom of fuze cavity counterbore	Visual
211.	Depth from bottom of small internal thread undercut to bottom of fuze cavity counterbore	Gage
212.	Depth from rear to bottom of large internal thread shoulder	Visual
213.	Length from front face to rear of external thread undercut	Gage
214.	Length, total	Gage
215.	Width of small thread undercut, max.	Gage
216.	Width of large external thread undercut	Gage
217.	Angle at front of fuze cavity missing or incorrect	Visual (see 6.1)
218.	Presence and location of wrench holes	Visual (see 6.1)
219.	Radii or chamfers missing or incorrect	Visual (see 6.1)
220.	Surface finish improper	Visual (see 6.1)
221.	Burr	Visual
222.	Foreign matter	Visual

4.3.2.8 Plug, Drawing C7549081, covering a detail of Drawing F7549103.

CRITICAL

None defined

MAJOR

101.	Pitch diameter of thread, min.	Gage
102.	Pitch diameter of thread, max.	Gage
103.	Major diameter of thread, min.	Gage
104.	Diameter at rear of dovetail, min.	Gage

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4.3.2.8 (Cont'd)

Defect	Method of Inspection
105. Angle of dovetail at rear of thread, missing or incorrect	Gage
106. Thickness from front face to start of dovetail	Gage
107. Metal defective	Visual
MINOR	
201. Diameter of counterbore	Gage
202. Depth of counterbore	Gage
203. Thickness	Gage
204. Presence and location of wrench holes	Visual (see 6.1)
205. Radii or chamfers, missing or incorrect	Visual (see 6.1)
206. Surface finish, improper	Visual (see 6.1)
207. Burr	Visual
208. Foreign matter	Visual

4.3.2.9 Metal Parts Assembly, Drawing F7549103 (prior to paint).

CRITICAL	
1. Metal defective	Visual (Probe Light)
MAJOR	
101. Runout of ogive with bourrelets	Gage
102. Runout of cone with bourrelets	Gage
103. Ogive overlapping body diameter	Gage
104. Wire slack between front of ogive and cone	Manual
105. Wire not properly sealed in conduit	Manual
106. Terminal not properly crimped	Manual
107. Adhesive on cone wall	Manual
108. Alignment of conduit hole on cone with slot on body	Visual
109. Length of wire from forward edge of rear undercut	Visual (see 6.1)
MINOR	
201. Surface finish improper	Visual (see 6.1)
202. Marking missing, incorrect or illegible	Visual
203. Burr	Visual
204. Foreign matter	Visual

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4.3.2.10 Metal Parts Assembly, Drawing F7549103 (after paint).

Defect	Method of Inspection
CRITICAL	
1. Metal defective	Visual (Probe Light)
MAJOR	
101. Weight	Approved Scale
102. Pitch diameter of chamber thread, max.	Gage
103. Pitch diameter of chamber thread, min.	Gage
104. Minor diameter of chamber thread, max.	Gage
105. Diameter of bourrelets, max.	Gage
106. Length, total, max.	Gage
107. Thread(s) damaged or not full	Visual
MINOR	
201. Cap, ogive not properly tightened	Manual
202. Protective coating not dry	Tactile/Manual
203. Protective coating inadequate or damaged	Visual
204. Burr	Visual
205. Foreign matter	Visual

4.3.2.11 Stop, Piston, Drawing C7549105, covering a detail of Drawing D7549111.

CRITICAL	
None defined	
MAJOR	
101. True position of large outside diameter with pitch diameter of thread and rear face	Gage
102. True position of angle rear, outside with pitch diameter of thread and rear face	Gage
103. Pitch diameter of thread, max.	Gage
104. Minor diameter of thread, max.	Gage
105. Outside diameter, large	Gage
106. Length to basic diameter	Gage
107. Thread damaged or not full	Visual
108. Metal defective	Visual

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4.3.2.11 (Cont'd)

Defect	Method of Inspection
MINOR	
201. Depth of wrench flats	Gage
202. Length, total	Gage
203. Location of wrench flats	Gage
204. Width across wrench flats	Gage
205. Radii or chamfers missing or incorrect	Visual (see 6.1)
206. Surface finish improper	Visual (see 6.1)
207. Protective coating, inadequate or damaged	Visual
208. Burr	Visual
209. Foreign matter	Visual

4.3.2.12 Piston, Drawing D7549107, covering a detail of Drawing D7549111.

CRITICAL	
1. Metal defective	Visual
MAJOR	
101. Runout of gear tooth with pitch diameter of thread	Gage
102. True position of outside diameter of body forward of shoulder with pitch diameter of thread and front of shoulder	Gage
103. Pitch diameter of thread, min.	Gage
104. Major diameter of thread, min.	Gage
105. Diameter of undercut at rear of thread	Gage
106. Diameter of body	Gage
107. Diameter of shoulder	Gage
108. Diameter of cavity	Gage
109. Diameter of neck, inside	Gage
110. Diameter of gas hole	Gage
111. Diameter of gear tooth	Gage
112. Location of basic diameter (inside)	Gage
113. Location of basic diameter (outside)	Gage
114. Location of gear tooth	Gage
115. Profile of gear tooth	Gage
116. Surface finish of body	Visual (see 6.1)
117. Thread(s) damaged or not full	Visual

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4.3.2.12 (Cont'd)

Defect	Method of Inspection
MINOR	
201. Diameter of neck, outside	Gage
202. Diameter of boss at rear of gear tooth	Gage
203. Depth to angle in neck cavity	Gage
204. Depth of slot in gear tooth boss	Gage
205. Length to shoulder	Gage
206. Length, total	Gage
207. Location of thread undercut	Gage
208. Width of thread undercut	Gage
209. Width of slot in gear tooth boss	Gage
210. Radii of chamfers missing or incorrect	Visual (see 6.1)
211. Surface finish, improper (except as otherwise classified)	Visual (see 6.1)
212. Protective coating inadequate or damaged	Visual
213. Burr	Visual
214. Foreign matter	Visual

4.3.2.13 Housing, Drawing D7549110, covering a detail of Drawing D7549111.

CRITICAL	
1. Metal Defective	Visual
MAJOR	
101. Runout of large diameter of groove with pitch diameter of chamber thread	Gage
102. Runout of large outside diameter with pitch diameter of chamber thread	Gage
103. True position of diameter of shear ring and small inside diameter with pitch diameter of chamber thread and boss seat	Gage
104. True position of counterbore with small inside diameter	Gage
105. Perpendicularity of piston seat with small inside diameters	Gage
106. Diameter of counterbore	Gage
107. Diameter of shear ring	Gage
108. Diameter inside (small)	Gage
109. Diameter of undercut (inside)	Gage

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4.3.2.13 (Cont'd)

	Defect	Method of Inspection
110.	Diameter of groove (large)	Gage
111.	Diameter of groove (small)	Gage
112.	Diameter, outside large	Gage
113.	Diameter of pivot pin holes	Gage
114.	Depth from front face to bottom of undercut (inside)	Gage
115.	Depth of groove	Gage
116.	Length of undercut (inside)	Gage
117.	Location of pivot pin holes from piston seat	Gage
118.	Location of pivot pin holes from centerline of housing	Gage
119.	Location of lugs	Gage
120.	Thickness of shear ring	Gage
121.	Width between lugs, for the minimum indicated depth	Gage
122.	Allignment of pivot pin holes to lugs	Gage
123.	Allignment of lugs	Gage
124.	Surface finish on inside of lugs	Visual (see 6.1)
125.	Surface finish of small inside of diameter rear of undercut	Visual (see 6.1)
126.	Surface finish of shear ring diameter	Visual (see 6.1)

MINOR

201.	Diameter outside (small)	Gage
202.	Diameter of chamber thread undercut	Gage
203.	Diameter of piston seat	Gage
204.	Diameter of base of lugs	Gage
205.	Depth from end of lug to piston seat	Gage
206.	Length from front face to chamber seat	Gage
207.	Length, total	Gage
208.	Thickness from chamber seat to bottom of fin slot	Gage
209.	Width across outside of lugs	Gage
210.	Radii or chamfers missing or incorrect	Visual (see 6.1)
211.	Surface finish improper (except as otherwise classified)	Visual (see 6.1)
212.	Protective coating inadequate or damaged	Visual
213.	Burr	Visual
214.	Foreign matter	Visual

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4.3.2.14 Pin, Fin, Drawing A7549106, covering a detail of Drawing D7549111.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. Diameter outside	Gage
102. Metal defective	Visual
MINOR	
201. Length, total	Gage
202. Radius missing or incorrect	Visual (see 6.1)
203. Protective coating, inadequate or damaged	Visual

4.3.2.15 Fin, Drawing D7549109, covering a detail of Drawing D7549111.

CRITICAL	
1. Metal defective	Visual
MAJOR	
101. Perpendicularity of pivot boss surfaces with pivot pin diameter	Gage
102. Perpendicularity of fin blade with pivot pin diameter	Gage
103. Diameter of pivot pin hole	Gage
104. Distance from centerline of pivot pin hole to top of 0.216 diameter measuring wire in gear tooth space	Gage
105. Location of pivot pin hole	Gage
106. Thickness through pivot boss	Gage
107. Angle of gear tooth space with pivot pin hole	Gage
108. Clearance angle at front end	Gage
109. Radii or chamfers, missing or incorrect	Visual (see 6.1)
110. Surface finish on pivot bosses	Visual (see 6.1)
MINOR	
201. Length, total	Gage
202. Width at front of fin blade	Gage
203. Width at rear of fin blade	Gage
204. Surface finish improper (except as otherwise classified)	Gage
205. Burr	Gage
206. Foreign matter	Gage

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4.3.2.16 Fin Assembly, Drawing D7549111.

Defect		Method of Inspection
CRITICAL		
	None defined	
MAJOR		
101.	Torque test (direction of assembly)	Torque Wrench
102.	Pitch diameter of housing thread, min.	Gage
103.	Pitch diameter of housing thread, max.	Gage
104.	Major diameter of housing thread, min.	Gage
105.	Assembly not operable under hand pressure	Manual
106.	"O" ring missing or damaged	Visual
MINOR		
201.	Diameter of threaded wall, inside	Gage
202.	Diameter across fin blades for minimum distance indicated	Gage
203.	Location of fin blades from centerline of assembly for minimum distance indicated	Gage
204.	Length, total, max.	Gage
205.	Pivot pins loose	Manual
206.	Threads damaged or not full	Visual
207.	Grease missing from piston wall	Visual
208.	Protective coating inadequate or damaged	Visual
209.	Marking missing, incorrect or illegible	Visual
210.	Burr	Visual
211.	Foreign matter	Visual

4.3.2.17 Carton, packed, prior to sealing.

CRITICAL		
	None defined	
MAJOR		
	None defined	
MINOR		
201.	Support missing or damaged	Visual
202.	Separator(s) missing or damaged	Visual
203.	Improperly packed	Visual

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4.3.2.18 Carton, packed, after sealing.

Defect		Method of Inspection
CRITICAL		
None defined		
MAJOR		
None defined		
MINOR		
201.	Contents loose	Manual
202.	Carton damaged, cut or gouged through all layers	Visual
203.	Carton not properly sealed	Visual
204.	Marking missing, incorrect or illegible	Visual

4.3.2.19 Pallet, Drawing F8837835.

CRITICAL		
None defined		
MAJOR		
None defined		
MINOR		
201.	Strapping missing, loose or broken	Visual/Manual
202.	Pallet damaged	Visual
203.	Improperly packed	Visual

4.3.3 Inspection testing. The following tests shall be performed in accordance with 4.4.

4.3.3.1 Mechanical properties.

4.3.3.1.1 Hardness.

4.3.3.1.1.1 Heat treatment. In order to determine uniformity of heat treatment, each body forging in the heat treatment lot (as defined by 4.3.1.3) shall be subjected to a hardness test with Rockwell hardness tester, or by any other method approved by the procuring activity (See 6.7).

4.3.3.1.1.2 Stress relief lots. Twenty-five stress relieved bodies from the first batch of each stress relief lot (as defined by 4.3.1.4) processed through the stress relief furnace and five bodies from each hour's production thereafter for the balance of the lot shall be subjected to a hardness test with a Rockwell hardness tester or by any other method approved by the procuring activity, to assess uniformity of the stress relief furnace. Hardness values shall be recorded.

4.3.3.1.2 Tension test.

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4.3.3.1.2.1 Heat treatment lots. Two projectile bodies, one of the hardest and one of the softest as determined by 4.3.3.1.1.1 shall be subjected to this test. Failure of any tension test specimen to comply with the yield strength or elongation requirements of the applicable drawing shall be cause for rejection of the lot. At the option of the supplier, a production control sample is used, the hardness range thus established shall be maintained throughout the heat treatment lot. The size of the sample shall be determined by the contractor. Projectile bodies which fall outside of the established range may be re-heat treated to fall within the range or new mechanical properties may be determined from newly selected projectile bodies which will represent the true extremes of the harndess in the heat treatment lot (See 6.7).

4.3.3.1.2.2 Stress relief lots. Two bodies, one of the hardest and one of the softest as determined by the hardness test from the first twenty-five bodies selected in accordance with 4.3.3.1.1.2 shall be subjected to this test. Failure of any tension test specimen to comply with the mechanical properties requirements of the applicable drawing shall be cause for rejection of the lot. When it has been determined that the mechanical properties comply and furnace temperature and time have been established, the furnace time and temperature shall be maintained throughout the stress relieved lot. The furnace temperature may vary within twenty-five degrees F., but not below the minimum temperature of 700 degrees F., and the soak time of 30 minutes minimum. Furnace temperature and time will be recorded every hour to assess adequate control of stress relief equipment. The furnace temperature will be calibrated every eight hours by an independent method.

4.3.3.1.2.3 Piston. After final heat treatment, two pistons from each lot shall be subjected to the tension test to determine compliance with the yield strength and elongation requirements of the applicable drawing. When the piston is machined from bar stock and provided mechanical properties are not altered in subsequent operations, tensile test specimens may be obtained from the mid radius position of the bar stock in lieu of obtaining the specimen from the piston after machining. Failure to comply with requirements shall be cause for rejection of the lot.

4.3.3.1.2.4 Aluminum components. The sample size and acceptance criteria shall be as specified in the applicable material specification, except that the mechanical properties requirements, as specified on the applicable component drawing shall apply. Failure to comply with the requirements of the applicable drawing and specifications shall be cause for rejection of the lot.

4.3.3.2 Hydrogen embrittlement. After final annealing, one copper cone from each lot shall be examined for hydrogen embrittlement. Failure to comply with requirements of 3.9.2 shall be cause for rejection of the lot represented by the sample, (see 6.4).

4.3.3 Wire assembly.

4.3.3.3.1 Strength of wire assembly. Prior to assembly into the metal parts assembly, ten wire assemblies from each lot shall be subjected to a pull test to determine compliance with 3.3.1.

4.3.3.3.2 Electrical continuity. Each metal parts assembly shall be subjected to this test to determine compliance with 3.3.2. Failure of any assemlby to comply shall be cause for rejection of the assembly.

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4.3.3.3.3 Fuze wire terminal. The fuze wire terminals of twenty-five metal parts assemblies from each lot shall be subjected to a pull test to determine compliance with 3.3.3. Failure of any terminal to comply with the requirements shall be cause for rejection of the lot of metal assemblies represented by the sample.

4.3.3.3.4 Insulation breakdown. Each metal parts assembly shall be subjected to this test to determine compliance with 3.3.4. Failure of any assembly to comply shall be eause for rejection of the assembly.

4.3.3.3.5 Electrical resistance. Each metal parts assembly shall be subjected to this test to determine compliance with 3.4. Failure of any assembly to comply shall be cause for rejection of the assembly.

4.3.3.4 Torque.

4.3.3.4.1 Body to ogive. Two assemblies from each 1 hours production or fraction thereof per each torque machine in operation shall be subjected to this test. Failure of any assemlby to comply with the requirement of the applicable drawing shall be cause for rejection of the production represented by the sample.

4.3.3.5 Cleaning and application of protective coating. The sample size and acceptance criteria shall be as specified in the applicable specification. Failure to comply with the requirements shall be cause for taking the corrective action prescribed herein.

4.3.4 Proving Ground Functional Tests. The following proving ground tests will be performed by the Government.

4.3.4.1 First article sample.

4.3.4.1.1. Initial production sample. The quantities listed in Table I shall be ballistically tested in the phases indicated.

TABLE I

Test phase	Quantity
Excess pressure	10
Service pressure	10
*Service pressure (Penetration)	10

*Penetration. Ten metal parts assemblies from the initial production sample only will be subjected to this test. Failure to comply with 3.8.c shall be cause for rejection of the initial production sample.

4.3.4.2 Production lot sample. The quantities listed in Table II shall be ballistically tested in the phases indicated.

TABLE II

Test phase	Quantity
Excess pressure	10
**Service pressure	10

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4.3.4.2 (Cont'd)

**After three consecutive production lots have been ballistically tested and accepted for this phase, ballistic testing may be discontinued with the approval of the procuring activity. However, implementation of ballistic testing will be instituted whenever directed by the procuring activity.

4.3.4.3 Excess pressure. Samples provided in accordance with Tables I and II shall be subjected to this test. Failure of any metal parts assembly to comply with the applicable requirements of 3.8 shall be cause for rejection of the initial production sample or the production lot represented by the sample.

4.3.4.4 Service pressure. Samples provided in accordance with Tables I and II shall be subjected to this test. Failure of any metal parts assembly to comply with the requirements of 3.8 shall be cause for rejection of the initial production sample or production lot represented by the sample.

4.3.5 Inspection equipment. Index of inspection equipment lots LI 7549103 and 7549111 identifies the inspection equipment required to perform the examination and tests prescribed herein. Equipment design shall be in accordance with the code designation of the applicable LI. (Reference is made to Inspection Equipment List for definition of coding format).

4.4 Test methods and procedures.

4.4.1 Mechanical properties. (See 6.5)

4.4.1.1 Tension test.

4.4.1.1.1 Bodies. From the samples selected in accordance with 4.3.3.1.2.1 or 4.3.3.1.2.2, two tension test specimens conforming to the largest obtainable standard "R" type specimen as prescribed by ASTM Method E-8 shall be taken longitudinally from opposite sides of the body 180 degrees apart. The number of specimens and their location is to determine the acceptability of the mechanical properties requirements and the Government shall not be restricted from testing additional specimens at other locations to determine that the mechanical properties of the projectile meet the drawing requirements. The criteria for performing this test shall be as prescribed by ASTM Method E-8, except the elongation shall be measured to the nearest thousandth (0.000) prior to conversion into percent elongation. Percent elongation shall be recorded to the nearest tenth (0.0) (See 6.7).

4.4.1.1.2 Piston. From the samples selected in accordance with 4.3.3.1.2.3 two tension test specimens as prescribed by ASEM Method E-8 shall be taken longitudinally from the opposite sides of the wall of each sample. The test shall be performed using the procedures and equipment prescribed by ASTM Method E-8.

4.4.1.1.3 Aluminum components. Procedures for performing in mechanical properties tests on aluminum components shall be in accordance with the criteria prescribed on the applicable material specification.

4.4.2 Hydrogen embrittlement. From the cone supplied in accordance with 4.3.3.2, two specimens, one from the front (large) end and one from the rear

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4.4.2 (cont'd)

(small) end, shall be removed for micro structure study. Specimens shall be polished, etched if desired, and examined for embrittlement under a microscope of not less than 75 diameters.

4.4.3 Wire assembly.

4.4.3.1 Strength of wire assembly. The samples selected in accordance with 4.3.3.3.1 shall be placed in a holding fixture and the load applied gradually in the direction of the wire axis. Assemblies tested shall not be used in metal parts assemblies.

4.4.3.2 Electrical continuity. The voltage shall be applied between the eyelet terminal and the fuze wire terminal and the conduit monitored for evidence of discontinuity.

4.4.3.3 Fuze wire terminal. The samples selected in accordance with 4.3.3.3.3 shall be tested using a suitable holding fixture and the load applied gradually to the terminal in the direction of the wire axis.

4.4.3.4 Insulation breakdown. The direct current shall be applied between the fuze terminal and the rear thread of the body.

4.4.3.5 Electrical resistance. The resistance between the forward end of the ogive and the rear thread of the body shall be checked with an OHM meter capable of performing this test accurately and consistently.

4.4.4 Torque.

4.4.4.1 Body to ogive. From the samples supplied in accordance with 4.3.3.2.1, the minimum torque required by the applicable drawing shall be applied in the direction of assembly, any perceptible movement will be cause for rejection.

4.4.5 Protective coating. The procedures for testing the cleaning of components and the application of protective coating shall be in accordance with the criteria for testing as specified in the applicable specification. The Salt Spray prescribed shall be performed in compliance with ASTM Method B-117.

4.4.6 Proving ground functional test.

4.4.6.1 Excess pressure. The ballistic test projectiles inert loaded and dummy fuzed will be fired in an acceptable standard gun based upon current gun tube condemnation criteria. A propellant charge adjusted to produce 109 to 117 percent of the upper pressure limit of the propellant will be used. Photographic equipemnt (smear camera) will be set up near the gun muzzle to record the flight of the projectile. Visual observation will be made to assure projectile impacts in target area. Velocity and chamber pressure will be recorded.

96

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4.4.6.2. Service pressure. The ballistic test projectiles inert loaded and dummy fuzed will be fired alternately with projectiles from a referenced lot for comparison. The projectiles will be fired in an acceptable standard gun based upon current gun tube condemnation criteria. The projectile shall be fired for accuracy at a range selected by the proof director. Photographic equipment (smear camera) will be set up near the gun muzzle to record the flight of the projectile. Visual observation will be made to assure projectile impacts in the target area. Velocity and chamber pressure will be recorded.

4.4.6.2.1 Penetration. The metal parts assemblies supplied in accordance with Table I, live loaded and fuzed will be assembled into complete rounds utilizing the service charge and fired alternately with projectiles from a reference lot for comparison. The rounds will be fired against a 6 inch thick homogeneous armor plate placed 300 feet from the gun muzzle at an obliquity of 60 degrees from the vertical. The projectile shall be fired in a gun for which the projectile is standard, the tube of which will have a minimum remaining service life of 50 percent based on current gun tube condemnation criteria.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packing.

5.2. Packing.

5.2.1 Level A. Military pack - Not applicable.

5.2.2 Level B. Limited military pack - Not applicable.

5.2.3 Level C. Minimum military pack - The metal parts for projectile, HEAT, M344A1, shall be packed as follows:

5.2.3.1 Metal parts assembly. Part No. 7549103 shall be packed in accordance with Drawing F8837835. As an alternative method, carton pack may be used in accordance with F7548319.

5.2.3.2 Fin assembly. Part No. 7549111 shall be packed in accordance with MIL-STD-109. Pack 24 assemblies in a 6 x 4 pattern, using interlocking separators.

5.2.3.3 Chamber. Part No. 75490080 shall be packed in accordance with MIL-STD-109. Pack 36 chambers in a carton in a 3 x 4 pattern in 3 layers, using interlocking separators.

5.2.3.4 Plug. Part No. 754081 shall be packed in accordance with MIL-STD-1169. Pack 648 plugs in a carton in a 6 x 6 pattern, 18 plugs to a cell, using interlocking separators.

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5.2.3.5 Shipping. When component parts from more than one lot are shipped as a carload or truck load, each lot must be kept separate and the division between lots clearly indicated.

5.2.3.6 Pallet shall be marked in accordance with Drawing F8838735. Carton shall be marked in accordance with Drawing F7548319.

5.2.3.7 Data card. Data card information shall be as specified in MIL-STD-1167.

6. NOTES

6.1 Visual examination qualifications. When compliance with the applicable requirements is in doubt as a result of visual examination this characteristic may be measured or gaged to determine acceptability.

6.2 Visual standards. Visual standards to be established at contractor's plant in accordance with 3.7.3.

6.3 Velocity-accuracy. Proving Ground tests for velocity-accuracy are not acceptance tests, but the results will be carefully analyzed by the Armament Command to determine quality of ammunition produced. Evidence of excessive variation in any phase of the proving ground tests will result in an immediate investigation by the Armament Command.

6.4 After initial production sample acceptance and after the contractor has presented evidence to the procuring agency concerning the adequacy of his furnace atmosphere used in annealing, the contractor may request that the testing for hydrogen embrittlement be eliminated. The decision regarding elimination of testing for hydrogen embrittlement will be made at the discretion of the procuring agency based on the results of initial production sample inspection and on the evidence presented by the contractor concerning his annealing equipment.

6.5 Alternative mechanical properties testing. Alternative procedures for mechanical properties testing may be used by the contractor when such testing and procedures provide as a minimum the quality assurance specified herein. Prior to applying the alternative procedures, the contractor shall describe them in a written proposal and shall demonstrate for the approval of the responsible technical agency that their effectiveness is equal to or better than the quality assurance provisions specified herein.

6.6 Ordering data. Procurement documents should specify the title, number, and date of this specification.

6.7 Optional procedure for heat lot identity. For a period of at least four months of production, the contractor shall maintain heat lot identity as per requirements of item specification. From the data generated during the four month period, the contractor shall establish a reliable hardness range of high and low values whcih meet the yield strength and elongation requirements of the applicable drawing. During this period and when the contractor changes to a hardness range identity, it behooves the contractor

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6.7 (Cont'd)

to monitor the chemistry of the heats of steel to assure compliance to the established hardness range. Once the hardness range has been established by the contractor, the following criteria for the heat treatment shall apply:

a. Heat treatment lots (definition). Each eight (8) hours of production of body forgings processed thru heat treatment furnaces shall be designated as a heat treatment lot. The heat treatment lot thus constituted shall be the basis for selection of the samples for the tension test hereinafter prescribed.

b. Heat treatment lot hardness test. Each body in the heat treatment lot as defined in (a) above shall be subjected to a Brinell Hardness Test to verify that each body falls with the established hardness range. Projectile bodies which fall outside the established range shall be rejected and may be re-heat treated to fall within the range.

c. Heat treatment lot tension test. Two bodies after hardness testing shall be randomly selected by Government personnel from each 8 hours production or fraction thereof and shall be subjected to this test. Failure of any tension test specimen to comply with the yield strength or elongation requirements of the applicable drawing shall be cause for immediate halt of production, and the following screening procedure shall be implemented. All projectiles in the production line processed forward of the hardness test shall be screened 100% for hardness and the results recorded. The two highest and the two lowest hardness readings of the screened projectiles shall be subjected to mechanical property testing. If the results are favorable, the contractor may proceed with production in accordance with his established range. If results indicate that the high or low samples do not meet the mechanical properties, the contractor shall establish a new hardness range to assure compliance with the mechanical properties as specified on the applicable drawing. This new hardness range shall be approved by the Government technical agency. The screened projectiles shall be considered suspect and removed to a bond area where mechanical properties testing will be conducted by the contractor with a reduced hardness range as per the recorded values. When the mechanical properties meet the requirements of the drawing at a reduced range, all suspect projectiles falling within this range can be released for further processing. All other projectiles that fall out of the range will be considered rejects.

d. Hardness tester. Each machine in operation shall be calibrated prior to start of each day's production.

6.8 Steel blooms, billets, slabs and bars.

6.8.1 It is the responsibility of the manufacturer of the projectile metal parts to select a fully-killed carbon steel of a grade compatible with the abilities of the equipment and the methods of processing the metal within the operations of the shops of the forger and machiner to produce the projectile to the dimensions, mechanical properties and metal integrity specified as an end item requirement.

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6.8.2 The supplier of the selected steel raw material is responsible only for furnishing a metal composition, grade, and type with dimensions, weight, soundness and freedom from metallurgical or mechanical defect limitations identifiable with the standard AISI grade and complying with the specific ASTM designation and tests selected by the projectile manufacturer.

6.8.3 In the application of carbon steel blooms, billets, and slabs, forging quality processing techniques adapted to use of the selected steel should be applied with the expectation that there will be variables in the metal cross-sectional dimensions in weight per unit length, in the hardenability variance from heat to heat, in the mechanical condition of the surface, and in the straightness of the raw material. Economic advantages will be realized if the process and equipment within the shops of the forger and the machiner have been engineered in terms of equipment sequence of operations, tooling designs, handling devices, and processing techniques having the capabilities of compensating for such variables.

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1315 - A290





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DOCUMENT IDENTIFIER AND TITLE
MIL-P-50308B(MU) Projectile, Heat, 106MM, M344A1 with Fin M8 Metal Parts Assy

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO.
	DATE

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